Week #8: [Muscles of the Distal Upper Extremity]

(Non-hyperlinked = Dr. Caitlin Cassidy Module)

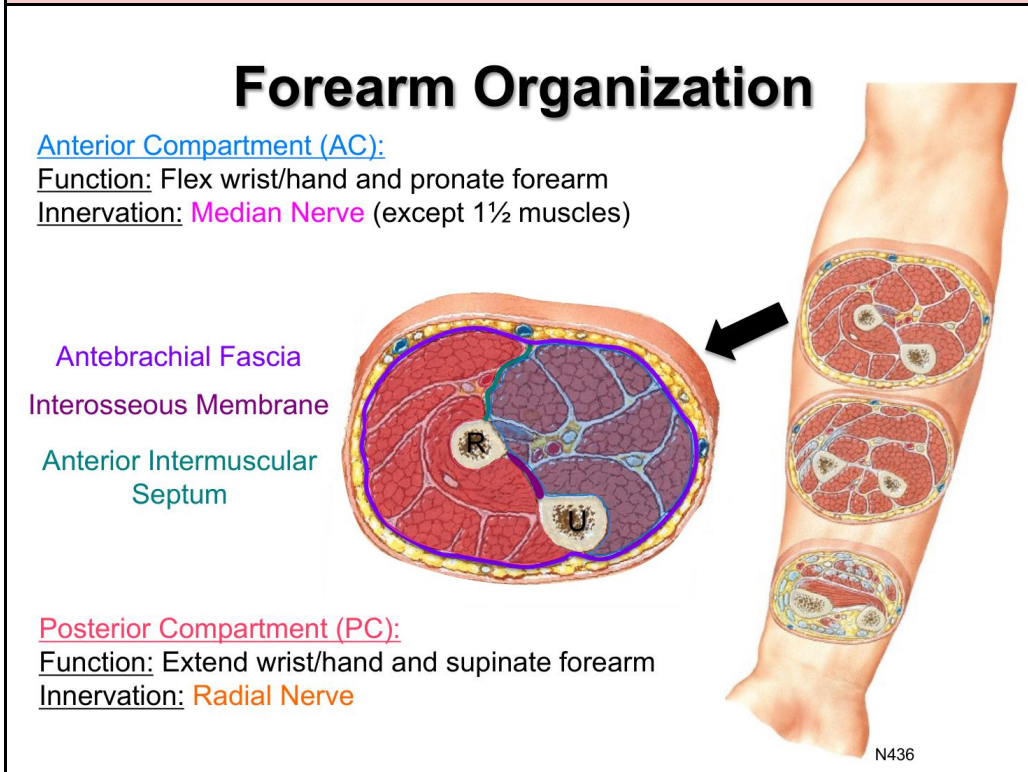
- Introduction to Module

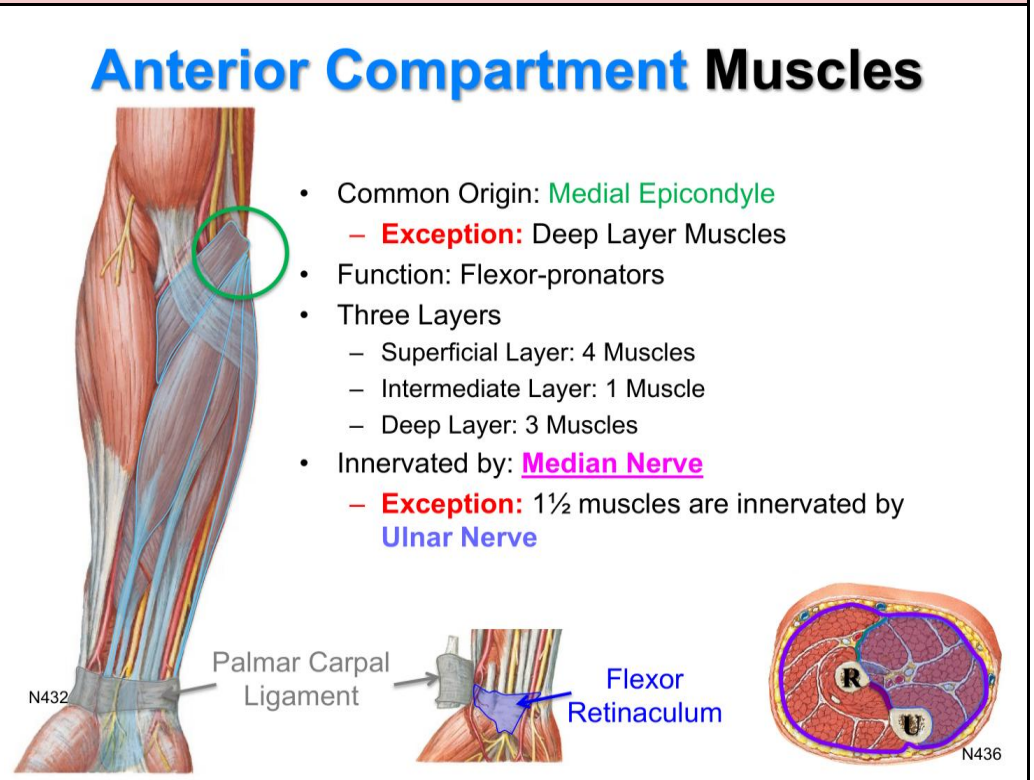
Muscles of the Forearm

- Muscles of the Anterior Forearm (43 min)
- Muscles of the Posterior Forearm (52 min)
- Approach to Muscle/Tendon Problems (19 min)

Muscles of the Anterior Forearm

- Anterior Compartment





I. Muscles:

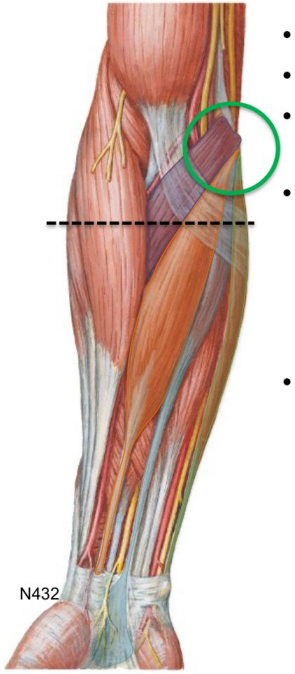
A. Forearm Organization: The forearm extends from the elbow to the wrist. Deep to the skin and subcutaneous tissue the deep fascia of the forearm, **antebrachial fascia,** is continuous with the brachial fascia. There are two bones in the forearm: the radius laterally and the ulna medially. These bones are connected via the **interosseous membrane.** There is one septum, the **anterior intermuscular septum**, extending from the radius to the antebrachial fascia. The bones, interosseous membrane and anterior intermuscular septum separate the forearm into anterior and posterior compartments.

a. **Anterior compartment** muscles function to flex the wrist and pronate the forearm. These muscles can be subdivided into three layers, the superficial layer has 4 muscles, the intermediate layer has 1 muscle, and the deep layer has 3 muscles. Many of these muscles share a common origin at the humeral medial epicondyle and they are all innervated by the ***median nerve*** with an exception of 1½ muscles, which are innervated by the ***ulnar nerve.***

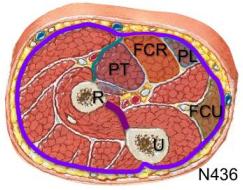
b. **Posterior compartment** muscles function to extend and supinate the wrist. These muscles can be subdivided into two layers, the superficial layer has 6 muscles and the deep layer has 5 muscles. Many of these muscles share a common origin at the ***humeral lateral epicondyle.*** Branches of the radial nerve innervate all muscles in this compartment.

i. **Ligaments** – The majority of the muscles in the forearm have tendons passing to the hand to act on the hand. To prevent bowstringing of the tendons during active contraction, ligaments surrounding the wrist adhere the tendons to the wrist. The palmar carpal ligament anteriorly is continuous with the extensor retinaculum posteriorly. The palmar carpal ligament often gets mistaken for the flexor retinaculum; however, the ***flexor retinaculum*** is deep and more distal to the palmar carpal ligament.

AC Muscles: Superficial Layer

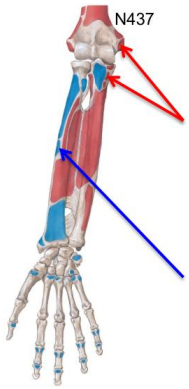


- Four Muscles
- Common Origin: Medial Epicondyle
- Pronator of wrist:
  - Pronator Teres
- Flexors of wrist:
  - Flexor Carpi Radialis
  - Palmaris Longus
  - Flexor Carpi Ulnaris
    - Innervated by Ulnar Nerve
- Common Innervation:
  - Median Nerve (with exception)



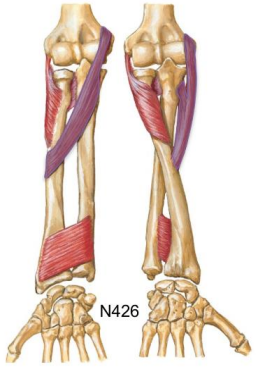


Pronator Teres

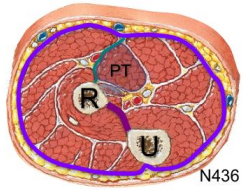


Origin:
Medial Epicondyle of Humerus
Ulnar Coronoid Process

Insertion:
Lateral Radius



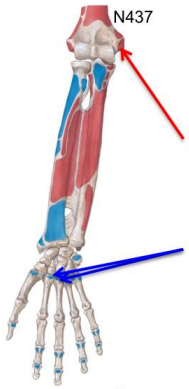
- Function:
  - Pronates Forearm
  - Elbow Flexion



- Innervation:
  - Median Nerve



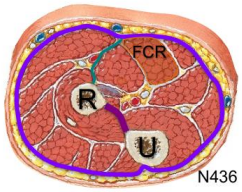
Flexor Carpi Radialis



Origin:
Medial Epicondyle of Humerus

Insertion:
Base of 2nd (3rd) Metacarpal

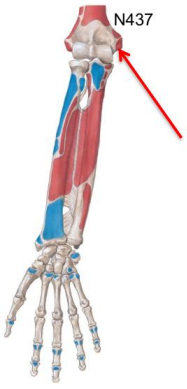
- Function:
  - Flexes Wrist
  - Abducts Wrist (Radial Deviation)



- Innervation:
  - Median Nerve

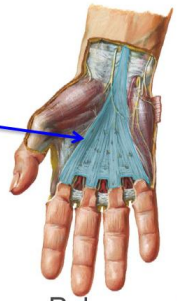


Palmaris Longus

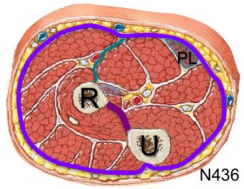


Origin:
Medial Epicondyle of Humerus

Insertion:
Palmar Aponeurosis


Palmar Aponeurosis

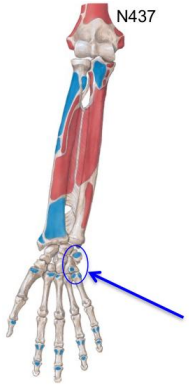
- Function:
  - Flexes Wrist
  - Tenses Palmar Aponeurosis



- Innervation:
  - Median Nerve



Flexor Carpi Ulnaris

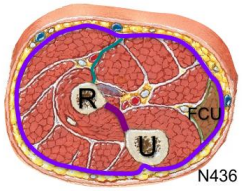


Origin:
Posterior Ulna



Insertion:
Wide insertion on medial carpals

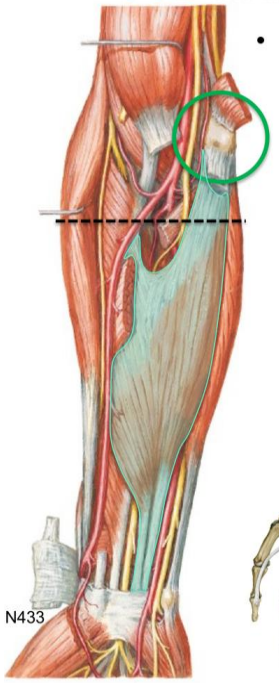
- Function:
  - Flexes Wrist
  - ADducts Wrist (Ulnar Deviation)

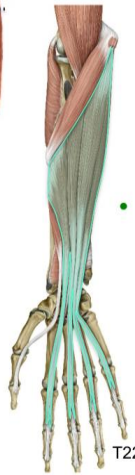


- Innervation:
  - Ulnar Nerve

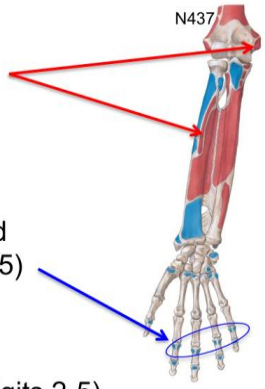


AC Muscles: Intermediate Layer





- One Muscle: Flexor Digitorum Superficialis

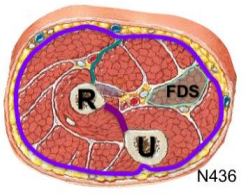


Origin:
Medial Epicondyle
Radius

Insertion:
Lateral Aspect of Mid Phalanges (Digits 2-5)

- Function:
  - Flexes MCP Joints (Digits 2-5)
  - Flexes Wrist

- Innervation:
  - Median Nerve



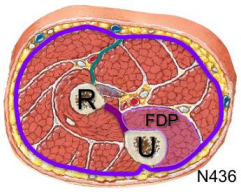
Anterior Compartment Muscles

Muscle	Origin	Insertion	Innervation	Function
Superficial Layer				
Pronator Teres	Humeral medial epicondyle & ulnar coronoid process	Lateral radius	Median Nerve	Pronates wrist & flexes elbow
Flexor Carpi Radialis	Humeral medial epicondyle	Base of 2nd (3rd) metacarpal		Flexes & abducts wrist
Palmaris Longus	Humeral medial epicondyle	Palmar Aponeurosis		Flexes wrist & tenses palmar aponeurosis
Flexor Carpi Ulnaris	Posterior ulna	Pisiform, hook of hamate, & 5th metacarpal	Ulnar Nerve	Flexes & adducts wrist
Intermediate Layer				
Flexor Digitorum Superficialis	Humeral medial epicondyle; oblique line of radius	Lateral aspect of middle phalanges of digits 2-5	Median Nerve	Flexes wrist & MCP jts of digits 2-5

AC Muscles: Deep Layer

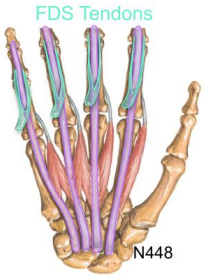


- Three Muscles
- Pronator:
  - Pronator Quadratus
- Flexors:
  - Flexor Pollicis Longus
  - Flexor Digitorum Profundus
- Common Innervation:
  - Median Nerve (with exception)



Flexor Digitorum Profundus



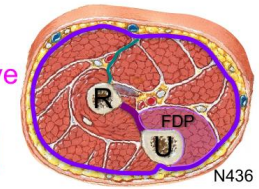


Origin:
Proximal ¾ of Ulna

Insertion:
Distal Phalanges (Digit 2-5)



- Function:
  - Flexes wrist
  - Flexes IP joints in Digit 2-5
- Innervation:
  - Lateral part (digits 2 & 3) – Median Nerve (anterior interosseous branch)
  - Medial part (digits 4 & 5) – Ulnar Nerve



Flexor Pollicis Longus




Flexion

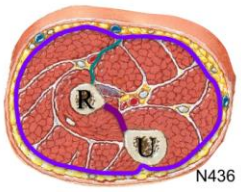
Origin:
Anterior Radius

Insertion:
Distal Phalanx of 1st Digit



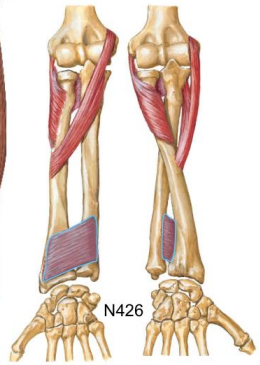
- Function:
  - Flexes wrist
  - Flexes MCP joint & IP joint in Digit 1
- Innervation:
  - Median Nerve (anterior interosseous branch)

Not in Cross-section



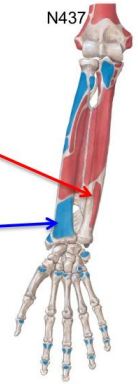
Pronator Quadratus





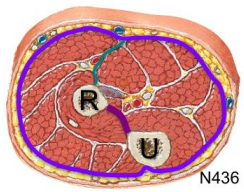
Origin:
Distal Ulna

Insertion:
Distal Radius



- Function:
  - Pronates Forearm
- Innervation:
  - Median Nerve (anterior interosseous branch)

Not in Cross-section



Deep Layer				
Flexor Digitorum Profundus	Anterior proximal ¾ of ulna	Distal phalanges of digits 2-5	Lateral part (to digits 2&3) – Median Nerve (anterior interosseous branch) Medial part (to digits 4&5) – Ulnar Nerve	Flexes wrist & Interphalangeal joints of digits 2-5
Flexor Pollicis Longus	Anterior radius	Distal phalanx of digit 1	Median Nerve (anterior interosseous branch)	Flexes wrist, MCP jt and Interphalangeal jt of digit 1
Pronator Quadratus	Distal ¼ of anterior ulna	Distal ¼ of anterior radius	Median Nerve (anterior interosseous branch)	Pronates forearm

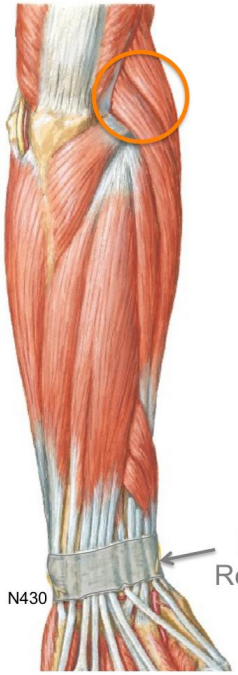
Clinical Note: **Medial Epicondylitis (Golfer's Elbow)** is an overuse injury of the common origin of the wrist flexors.

- Excessive wrist flexion may result in micro tearing of the common tendon leading to pain and inflammation.
- Medial epicondylitis is common in golfers due to the stress the golf swing puts on the common flexor origin. Treatment of medial epicondylitis is typically conservative with rest, ice, compression, and a stretch and strengthening regimen

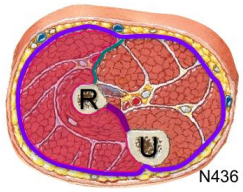
Muscles of the Posterior Forearm

- Posterior Compartment

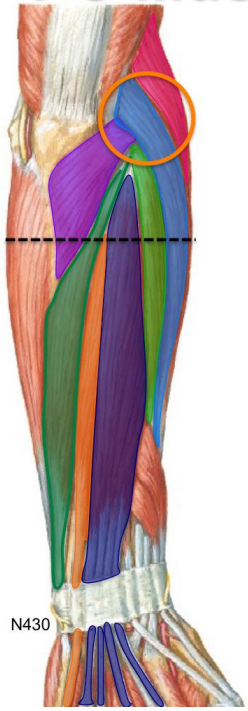
Posterior Compartment Muscles



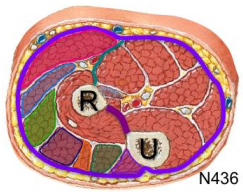
- Common Origin: Lateral Epicondyle
  - **Exception:** Deep Layer Muscles
- Function: Extensor-supinators
- TWO Layers
  - Superficial Layer: 6 Muscles
  - Deep Layer: 5 Muscles
- Innervated by branches of the Radial Nerve



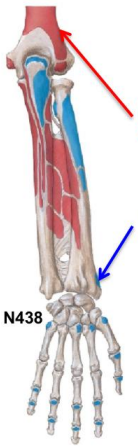
PC Muscles: Superficial Layer



- Six Muscles
- Common Origin: Lateral Epicondyle
- Non Extensor-supinators:
  - Anconeus
  - Brachioradialis
- Extensors:
  - Extensor Carpi Radialis Longus
  - Extensor Carpi Radialis Brevis
  - Extensor Digitorum
  - Extensor Digiti Minimi
  - Extensor Carpi Ulnaris
- Common Innervation:
  - Radial Nerve
    - Deep Radial Nerve



Brachioradialis



Origin: **Supra-epicondylar ridge of humerus**

Insertion: **Lateral surface of distal radius (proximal to styloid process)**



- Function:
  - Weakly flexes the forearm
    - Maximal strength in the mid pronated position
  - "drinking muscle"

Extensor Carpi Radialis Longus & Brevis



Origin:
Lateral Supraepicondylar Ridge (ECRL)
Lateral Epicondyle of Humerus (ECRB)

Insertion:
Base of 2nd MC (ECRL)
Base of 3rd MC (ECRB)

- Function:
  - Extends and Abducts wrist
  - ECRL active during fist clenching



Extensor Digitorum & Extensor Digiti Minimi

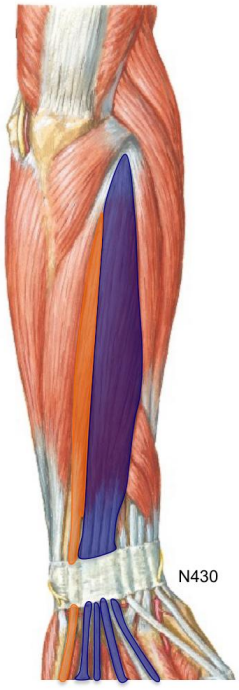


Origin:
Lateral Epicondyle of Humerus

Insertion: Extensor Expansion



- Function:
  - Extend MCP joints primarily
  - Extend IP joints secondarily



Extensor Carpi Ulnaris



Origin:
Lateral Epicondyle of Humerus
Post. Border of Ulna

Insertion: Base of 5th MC

- Function:
  - Extends and Adducts Wrist
  - Active during fist clenching



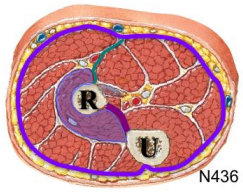
C. Posterior Compartment Muscles:

Muscle	Origin	Insertion	Innervation	Function
Anconeus	Humeral lateral epicondyle	Lateral olecranon		Weak forearm extension; elbow stabilization
Superficial Layer				
Brachioradialis	Humeral lateral supraepicondylar ridge	Lateral distal radius	Radial Nerve	Weakly flexes forearm (maximally in mid-prone position)
Extensor Carpi Radialis Longus	Humeral lateral supraepicondylar ridge	Base of 2nd metacarpal		Extends and a**b**ducts wrist; active during fist clenching
Extensor Carpi Radialis Brevis	Humeral lateral epicondyle	Base of 3rd metacarpal	Deep Branch of Radial Nerve	Extends and a**b**ducts the wrist
Extensor Digitorum	Humeral lateral epicondyle	Extensor expansions of digits 2-5		Extends MCP jt and PIP joints of digits 2-5
Extensor Digiti Minimi	Humeral lateral epicondyle	Extensor expansion of the 5th digit		Extends the 5th digit at the MCP and IP joints
Extensor Carpi Ulnaris	Humeral lateral epicondyle; posterior border of ulna	Base of 5th metacarpal		Extends and a**d**ducts the wrist; active during fist clenching
Deep Layer				
Supinator	Humeral lateral epicondyle; posterolateral border of ulna	Posterolateral and anterior surface of proximal ⅓ of radius	Deep Branch of Radial Nerve	Supination of forearm
Extensor Indicis	Posterior surface of distal ⅓ of ulna	Extensor expansion of the 2nd digit		Extends 2nd digit

PC Muscles: Deep Layer



- Five Muscles
- Supinator(s):
  - Supinator
- Extensor(s):
  - Extensor Indicis
- Outcropping Muscles
  - Abductor Pollicis Longus
  - Extensor Pollicis Longus
  - Extensor Pollicis Brevis
- Common Innervation:
  - Radial Nerve
    - Deep branch of Radial Nerve
    - Posterior Interosseous Nerve

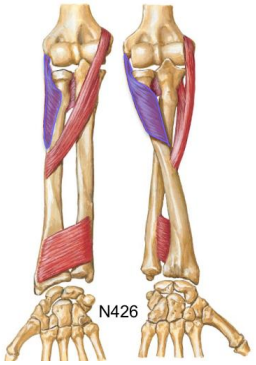


Supinator



Origin:
Lateral Epicondyle of Humerus
Posterolateral Border of Ulna

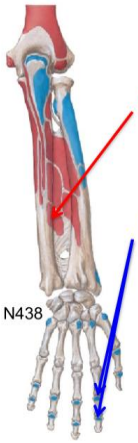
Insertion:
Posterior, Lateral, and Anterior Surface of Proximal ⅓ Radius





- Function:
  - Supinates Forearm

Extensor Indicis Proprius



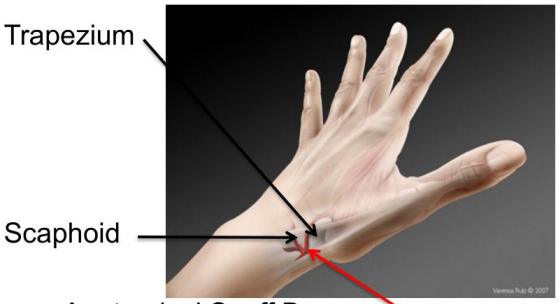
Origin:
Post. Surface of Distal ⅓ of Ulna

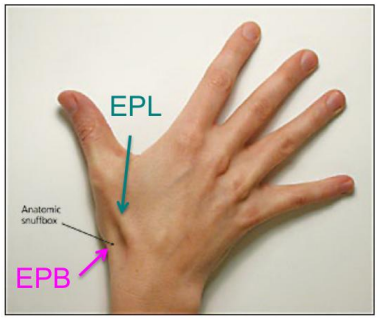
Insertion:
Extensor Expansion of 2nd Digit



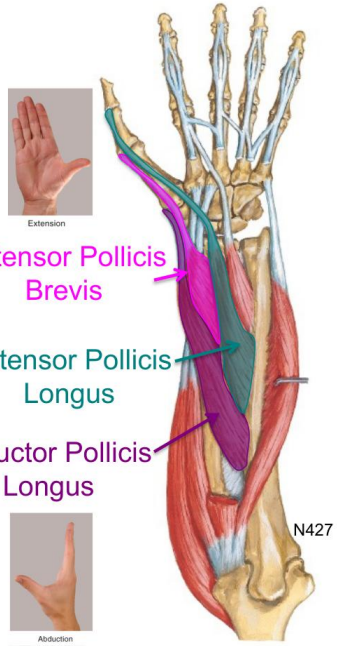
- Function:
  - Extends 2nd Digit
    - Enabling independent extension

Outcropping Muscles





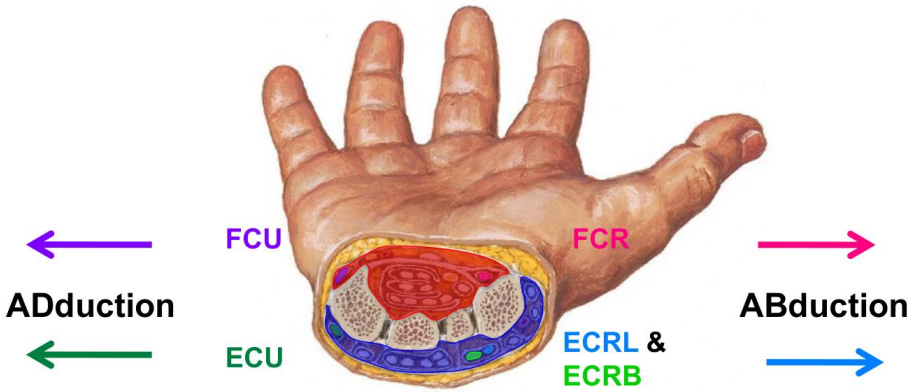




Abductor Pollicis Longus (outcropping muscle)	Posterior surface of proximal ½ of ulna, radius, and interosseous membrane	Base of 1st metacarpal	Posterior Interosseous Nerve	**Ab**ducts 1st digit; extends the CMC joint of 1st digit
Extensor Pollicis Longus (outcropping muscle)	Posterior surface of mid ⅓ of ulna and interosseous membrane	Base of distal phalanx of 1st digit		Extends IP joint of 1st digit; extends MCP and CMC joints
Extensor Pollicis Brevis (outcropping muscle)	Posterior surface of the distal ⅓ of radius and interosseous membrane	Base of proximal phalanx of 1st digit		Extends MCP joint of 1st digit; extends CMC joint of 1st digit

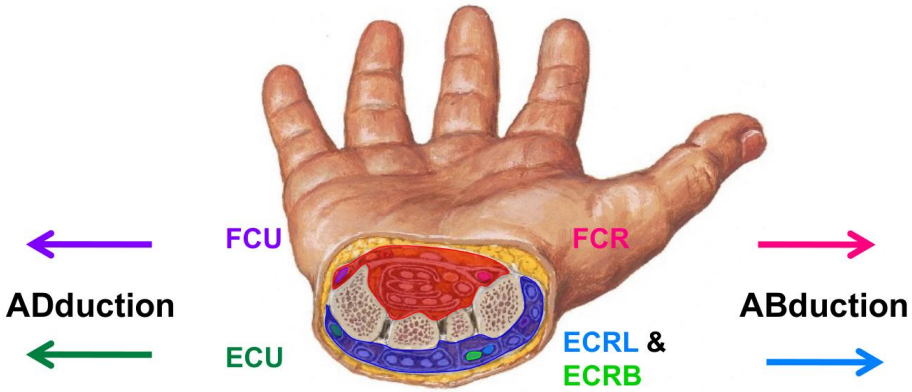
Movements of the Wrist

- Flexion – Anterior Compartment
- Extension – Posterior Compartment
- Abduction (Radial Deviation) – Both (FCR, ECRL & ECRB)
- Adduction (Ulnar Deviation) – Both (FCU & ECU)

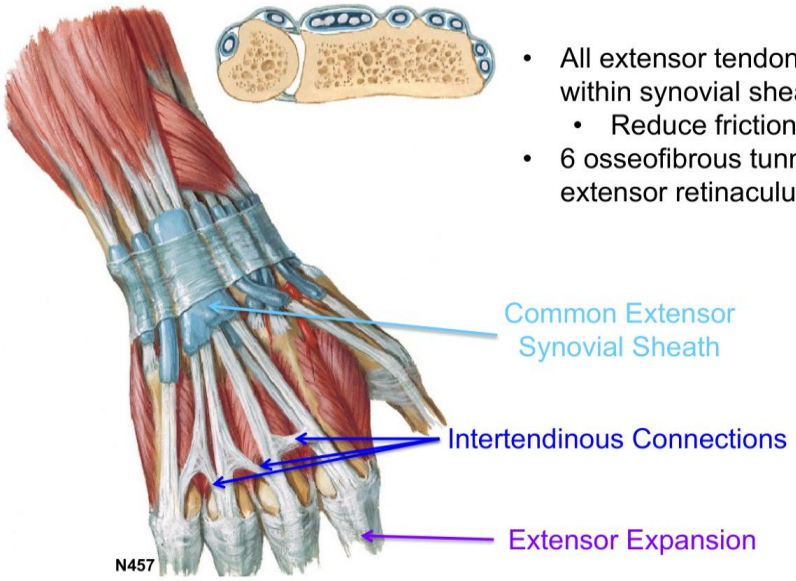


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Extensor Digitorum Features



- All extensor tendons wrapped within synovial sheaths
  - Reduce friction
- 6 osseofibrous tunnels deep to extensor retinaculum

Note: The **extensor expansion** is a flattening of the tendons of the extensor digitorum forming a triangular aponeurosis that wraps around the dorsum and sides of the metacarpals and proximal phalanges of each digit. Extensor digitorum, extensor digiti minimi, extensor indicis as well as the lumbricals and interossei insert via the extensor expansion.

Note: Each of the extensor tendons that cross the wrist joint is encased within a **synovial sheath,** to decrease friction, before passing under the extensor retinaculum. The space deep to the extensor retinaculum is divided into six osseofibrous tunnels formed by attachments of the extensor retinaculum to the ulna and radius.

Clinical Note: Lateral Epicondylitis (Tennis Elbow) – Inflammation of the lateral epicondyle usually resulting from overuse of the superficial extensor muscles, which originate on the lateral epicondyle. Pain is felt over the lateral elbow and radiates down the posterior forearm. Repeated forceful flexion and extension of the wrist strain the attachment of the common extensor tendon, producing inflammation of the periosteum of the lateral epicondyle.

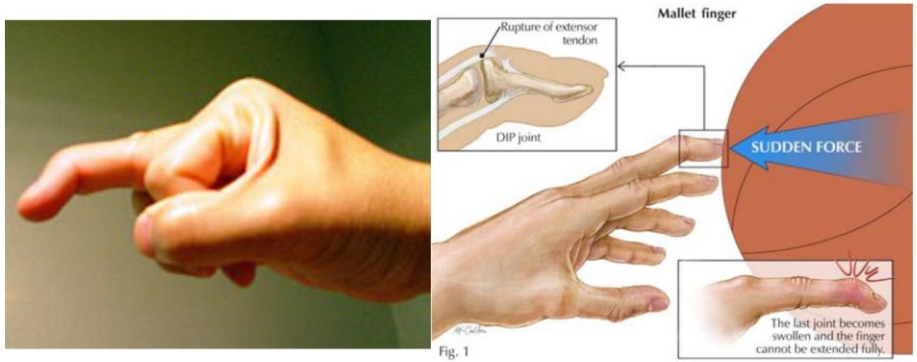
Synovial Cyst

- A cyst containing clear, mucinous fluid sometimes appears on the hand, usually the dorsum of the wrist.
- "*ganglion cyst*"
- Communicate with the synovial sheaths
- "Bible Bump"
- Treatment:
  - Nothing
  - Drainage or Excision



Clinical Note: Synovial Cyst of the Wrist– A cyst containing clear, mucinous fluid sometimes appears on the hand, most commonly on the dorsum of the wrist. It is referred to as a *"ganglion cyst"*. They are close to and often connected with the synovial sheaths of the extensor tendons.

Mallet Finger (Baseball Finger)



- Inability to extend the finger at the DIP joint.
- Results from partial tearing of the extensor tendon from its insertion into the base of the distal phalanx.

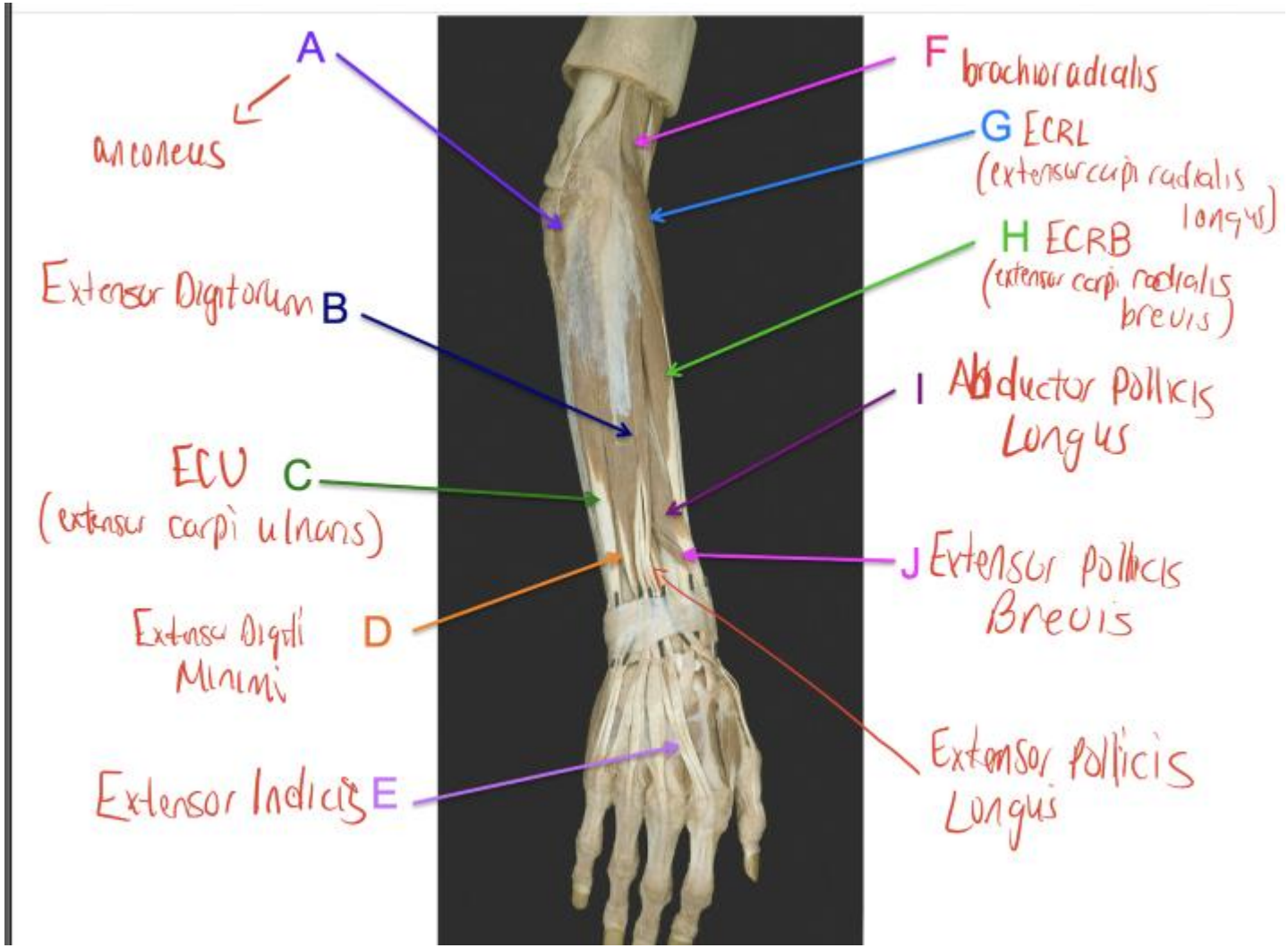
Treatment of Mallet Finger

- The most common treatment is RICE and splinting.
- If it is a complete avulsion fracture, surgery is required to pin the bone back down.





Clinical Note: Mallet Finger (Baseball Finger): Results from the partial tearing of the tendon of the extensor digitorum from its insertion on the base of the distal phalanx. Sudden hyperflexion of the finger, as occurs when jamming a finger, places too much strain on the extensor tendon causing it to avulse (slightly tear away). As a result the person is unable to extend the distal IP joint.



Approach to Muscle/Tendon Problems

Clinical Approach

History	Physical	Management Approach
• Look for patterns!! • OLDCARS • **Acute** (traumatic, inflammatory) vs **chronic** (degenerative, overuse) • Overlay your knowledge of anatomy (which structures are medial, which are lateral, etc)	• LOOK at it • Is it swollen? Are there obvious deformities? • FEEL it • Where does it hurt to touch? Does it feel hot? • MOVE it • Is ROM restricted? • Specific testing maneuvers for specific structures • Don't forget the nerves!	• What (if any) investigations will help you? • If no red flags, apply your pattern • Educate • Non-pharmacologic • Pharmacologic • Interventional • Remember your patient is a person!

Differential Diagnosis for Elbow Problems

By Anatomic Structure	By Location
• Bone/ligament/joint • Arthritis • Degen, post-traumatic, inflammatory • Subluxation/dislocation • Sprain/tear • UCL • Fracture • Muscle/tendon • Tendinopathies • Medial epicondylosis (golfer's elbow) • Lateral epicondylosis (tennis elbow) • Distal biceps • Distal triceps • Other structures • Olecranon bursitis • Ulnar neuritis/neuropathy	• Anterior • Arthritis • Degenerative, post-traumatic • Inflammatory • Distal bicep tendinopathy • Medial • Medial epicondylosis • Ulnar neuritis/neuropathy • UCL sprain/tear • Lateral • Lateral epicondylosis • Posterior • Olecranon bursitis • Arthritis • Triceps tendinopathy (fracture could be anywhere)

Muscle/Tendon Problems

- Issues with these anatomic structures may impact
  - Elbow pain
  - Elbow strength

Tendin**itis** vs Tendin**osis**:

(Does the undifferentiated doc really care? – *YES!*)

"100+ days" to make new collagen

Tendinitis	Tendinosis
Less common than we think	More common than we think
True inflammation from a variety of causes • Rheumatologic conditions • Acute injury (tendon overloaded too much or too suddenly) • Infection	Degeneration of collagen in tendon – not related to inflammatory cascade typically! • Chronic overuse with insufficient recovery time • Under the microscope, see shift to more immature and disorganized collagen, abnormal vascularity
Treatment: reduce inflammation • RICE for acute injuries • Anti-inflammatories • Treat underlying condition	Treatment: no inflammation typically to reduce! • Therapies to encourage collagen recovery • Analgesics (may still include anti-inflammatories)
Faster recovery if underlying problem	Slooooooooow recovery

Lateral Epicondylitis/osis

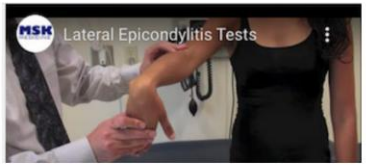
- Common: 1-3% of population annually
  - 30's and 40's
  - Occupational > recreational
  - "Tennis Elbow"
- Tendinopathy of extensor carpi radialis brevis predominantly (but could be other muscles from common extensor origin)
- Symptoms: pain at site
  - Just distal to lateral epicondyle
- Classic "pain and weakness"
  - Esp gripping + supination

Lateral Epicondylitis/osis

- Pain with palpation over common extensor tendon
- Pain with resisted extension middle finger +/-wrist extension +/-supination
- Pain with passive stretch of common extensor origin

Pain here





Medial Epicondylitis/osis

- Tendinopathy at common *flexor* origin
- Clinical
  - Pain at common flexor origin
  - Pain with resisted wrist flexion, pronation
  - Pain with passive stretch





Biceps Tendinopathy

	Tendon -itis / -osis	Tendon rupture
Common clinical points	• Insidious onset • Pain on resistance • Pain on stretch • Usual treatments	• Trauma or end-overuse • "Pop" heard or felt • Pain (no pain?) • Bruise • Weakness • Change in muscle
Proximal (long or short head)	Anterior shoulder pain	Not uncommon (esp i Often degenerative Usually still functiona non-op management
Distal biceps	Anterior elbow pain	Rare but consequential - poor supination, weak flexion Call: 1-800-ORTHO - **limited time offer**





Photo credits: https://www.youtube.com/watch?v=P6mWlpWfVfc, https://coreem.net/core/biceps-and-triceps-tendon-rupture/

Tendinitis/osis



- Management (from Approach to Management of ALL THINGS MSK
  - Education
    - Tell them what it is!
    - What's expected? Many will improve quickly, some will be quite chronic – address the offending movements/activities!!!
  - Non-pharm
    - Modalities (ice), RICE
    - Modify offending activity
      - Rest – either by time or by task
        - Adjust ergonomics/biomechanics – positioning, reduce excess force
    - Orthoses, taping – support and realign forces
      - Tennis elbow counterforce strap during offending activities
    - PT – eccentric strengthening BID x 12 weeks (stimulate collagen production/strength)
    - Massage – may stimulate fibroblasts and generate collagen
  - Pharm
    - Anti-inflammatories can be considered, but may not be as useful for chronic tendinOSIS (topical preferred)
  - Intervention/surgery
    - ?Regenerative options (PRP)
    - Surgical referral if refractory

Practice Pointer

- Could ask: "have you tried any physio?"
- Answer: "yes"

- Better "show me what you've been doing in physio"

1. Answer "uhhh,…"
2. Or hopefully:

